



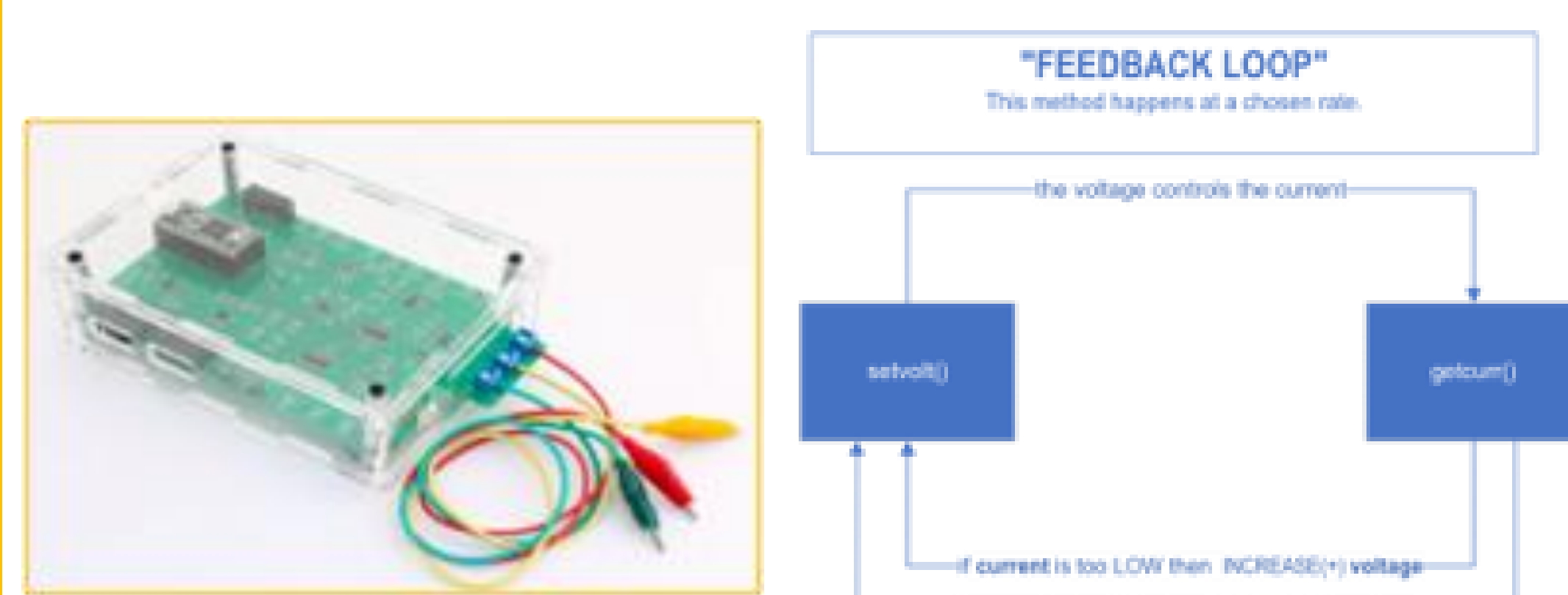
APPLYING GALVANOSTATIC METHODS TO IO RODEO POTENTIOSTAT

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INTRODUCTION

Energy is generated every day through natural gases, coal, nuclear energy and renewable energy sources such as solar, wind, etc. Where we store that energy matters. Batteries need to be able to hold energy and be able to deliver it to a system efficiently. In making better batteries research is done to examine battery materials and chemistry. Galvanostatic methods are used to measure the rate of an electrochemical reaction, applying constant current and measuring the voltage. My goal is to write software through Python API on the Teensy 3.2 development platform to allow the IO Rodeo potentiostat which only runs voltammetric tests to perform galvanostatic test by utilizing a feedback loop to be able to control current and measure voltage. The IO Rodeo is a portable and compact system that performs many voltammetric tests on chemical cells utilizing a three-electrode system. It can supply a set voltage to a cell and record the current passing through a cell.



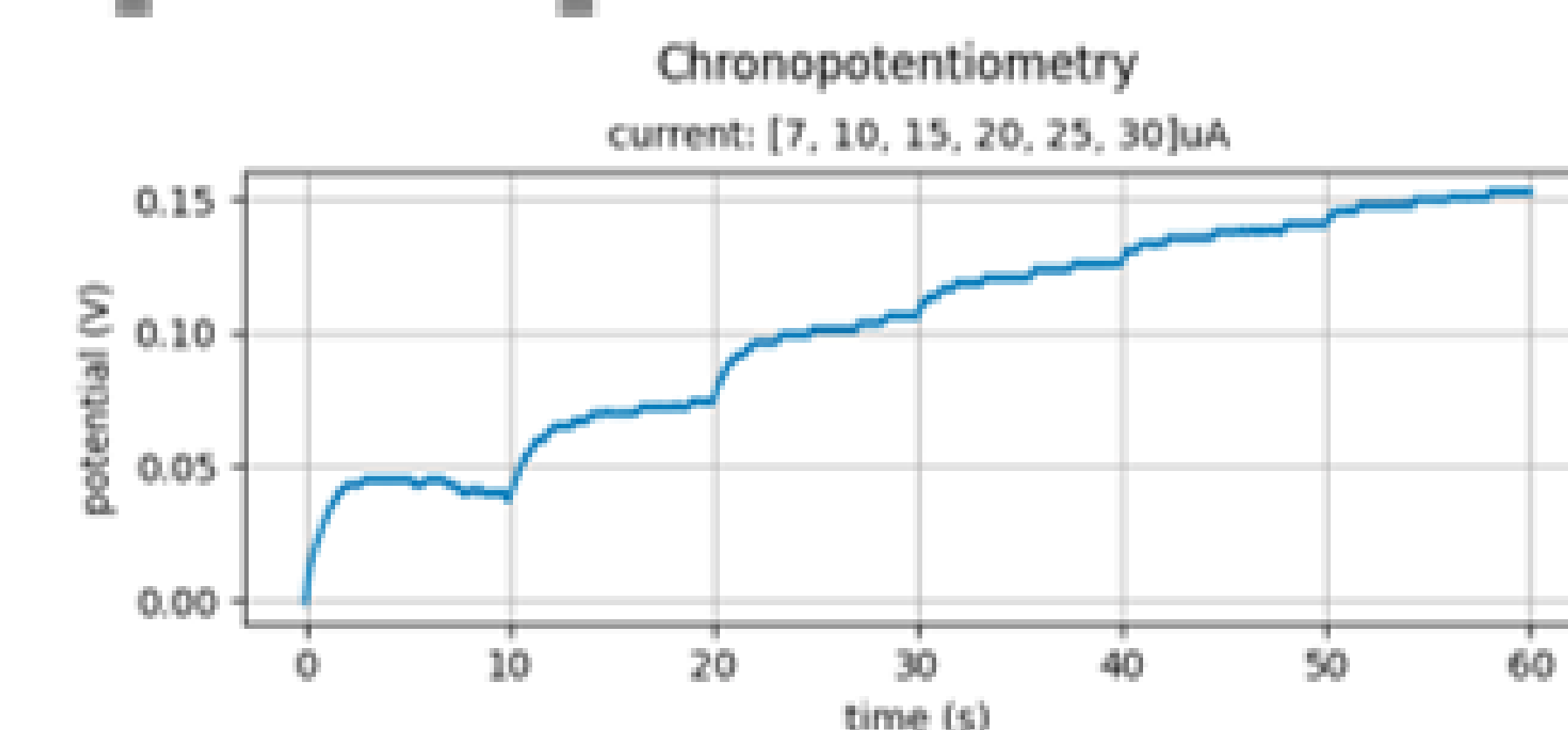
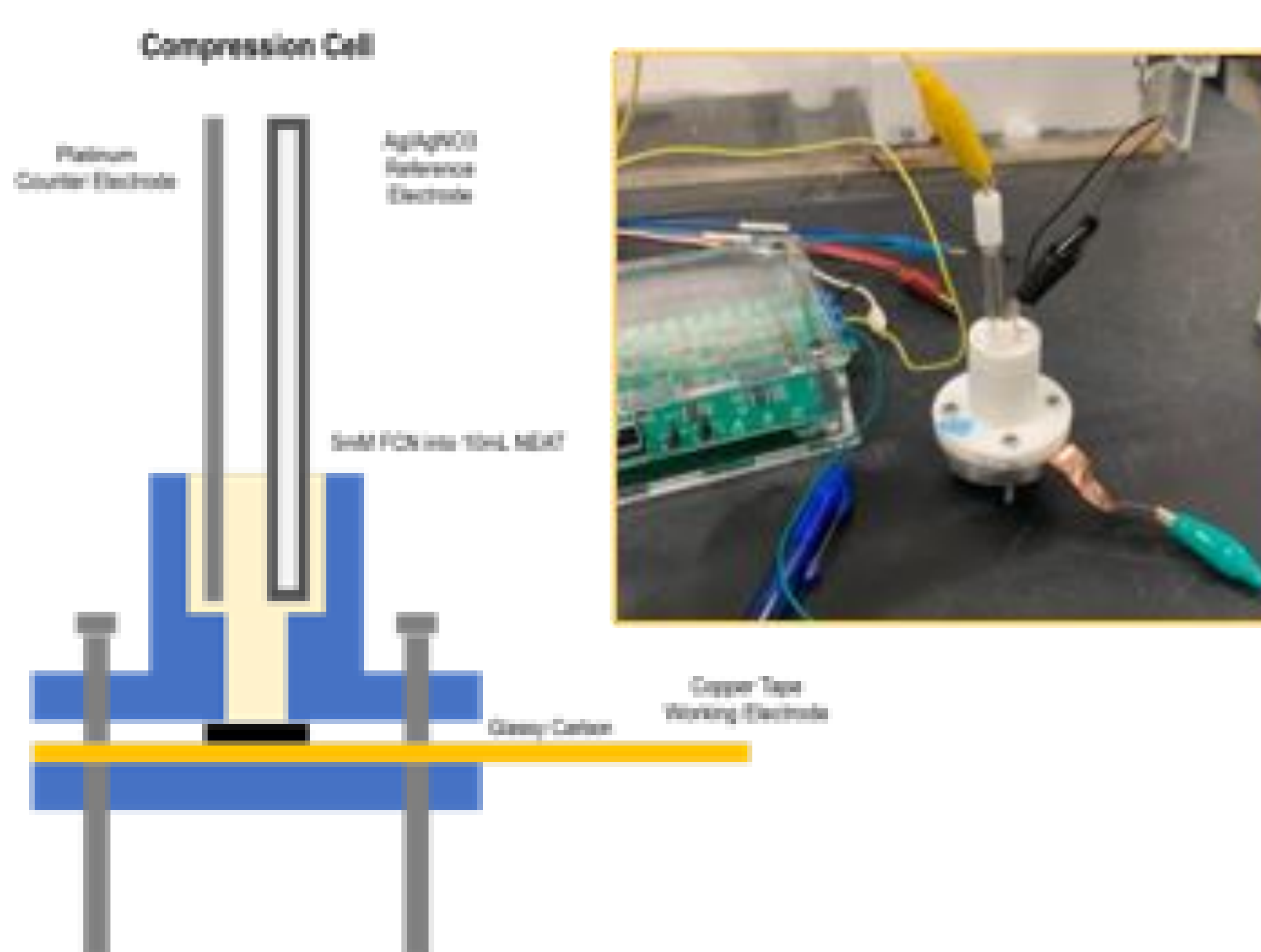
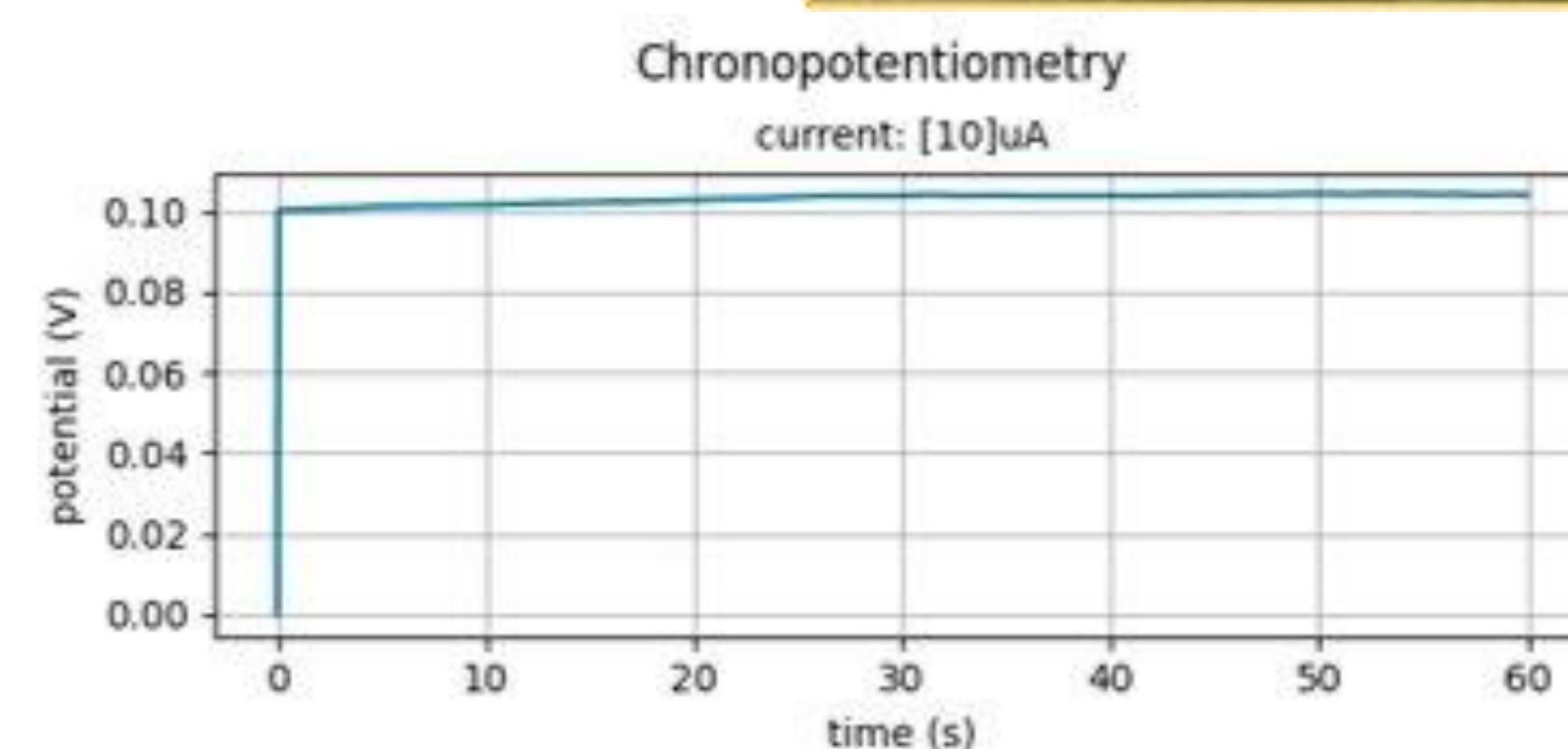
MOTIVATION

The IO Rodeo Potentiostat's compact size and low price make it an ideal system to do practice test with and produce qualitative results. It starts at \$200 compared to \$10,000 and it can be carried with a single hand whereas other systems require cart to transport. It is also an open-source system which makes it easily programmable.

CHRONOPOTENTIOMETRY (CP)

Chronopotentiometry is a galvanostatic method I applied to an electrochemical cell.

- Dummy Cell at 10 kilohms resistance at 10uAmps.

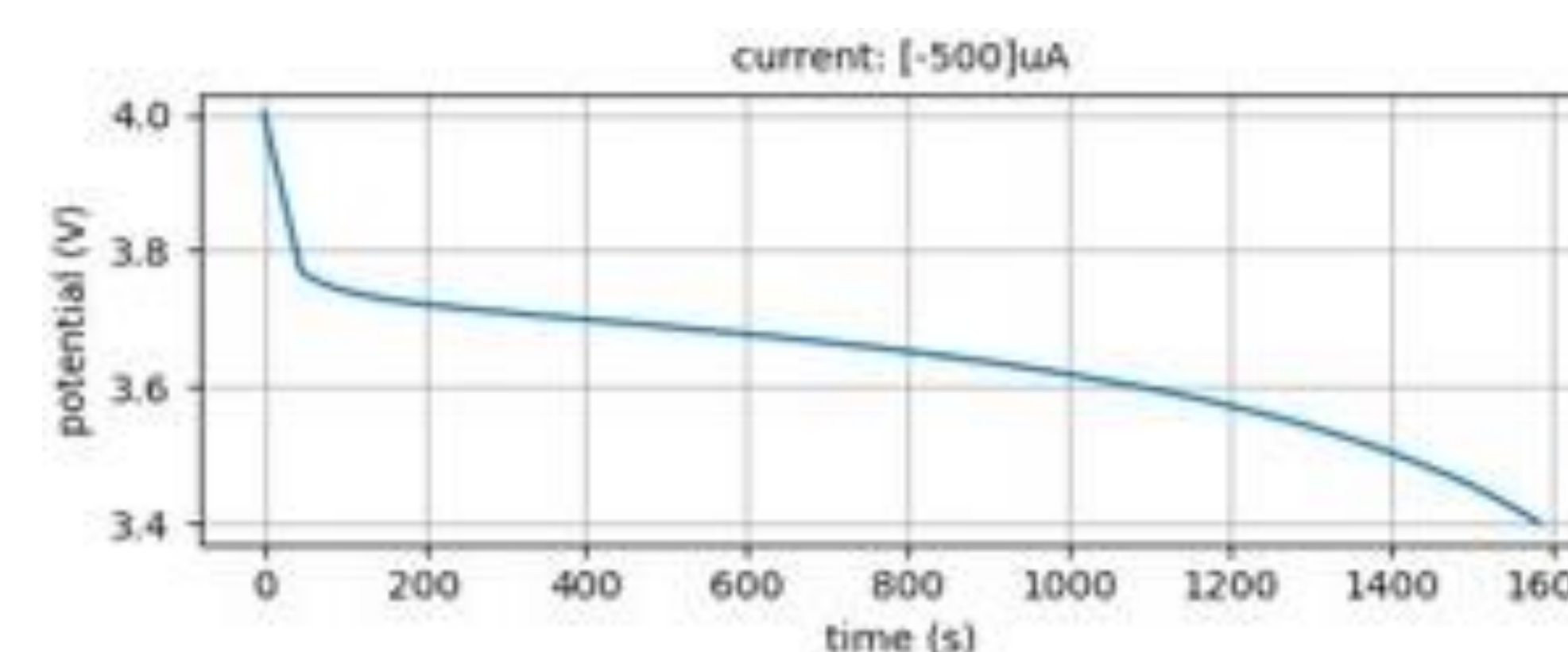
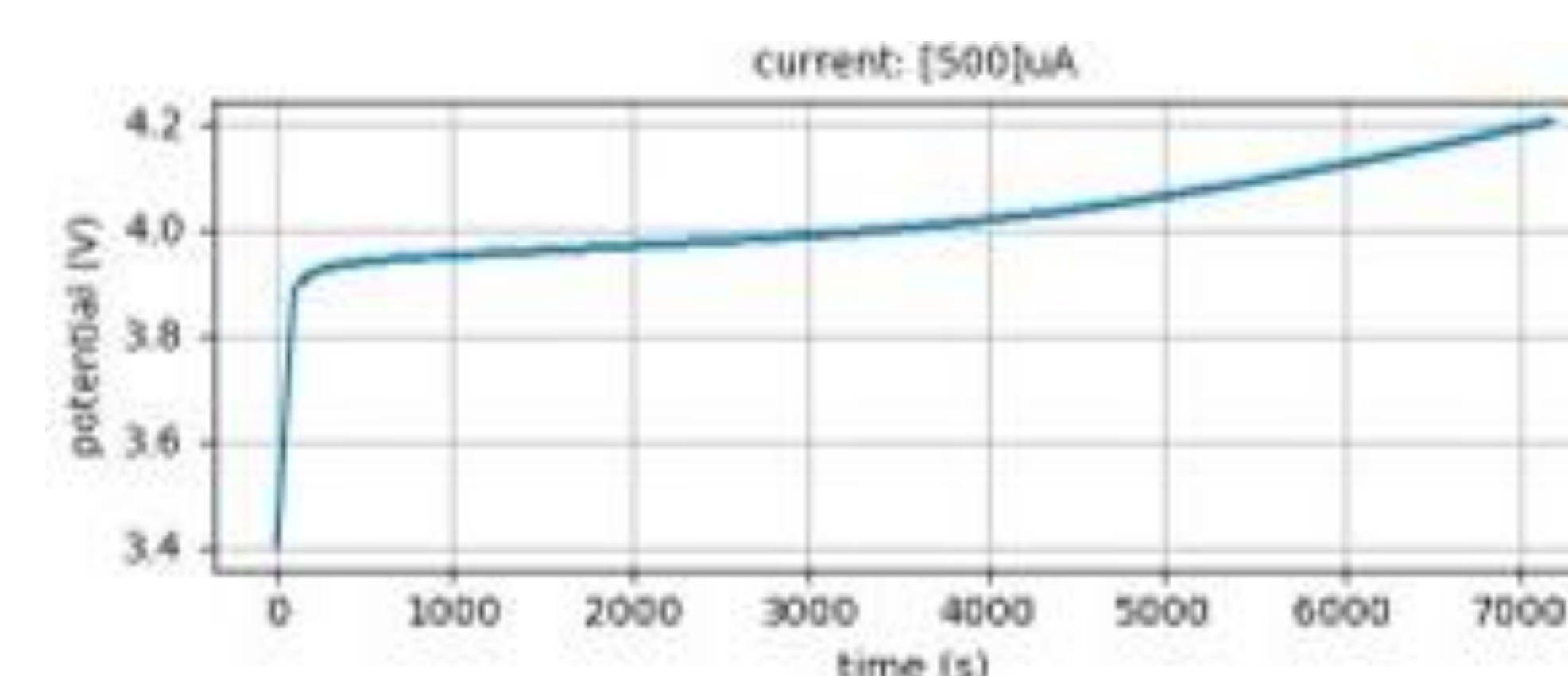


BATTERY CYCLING

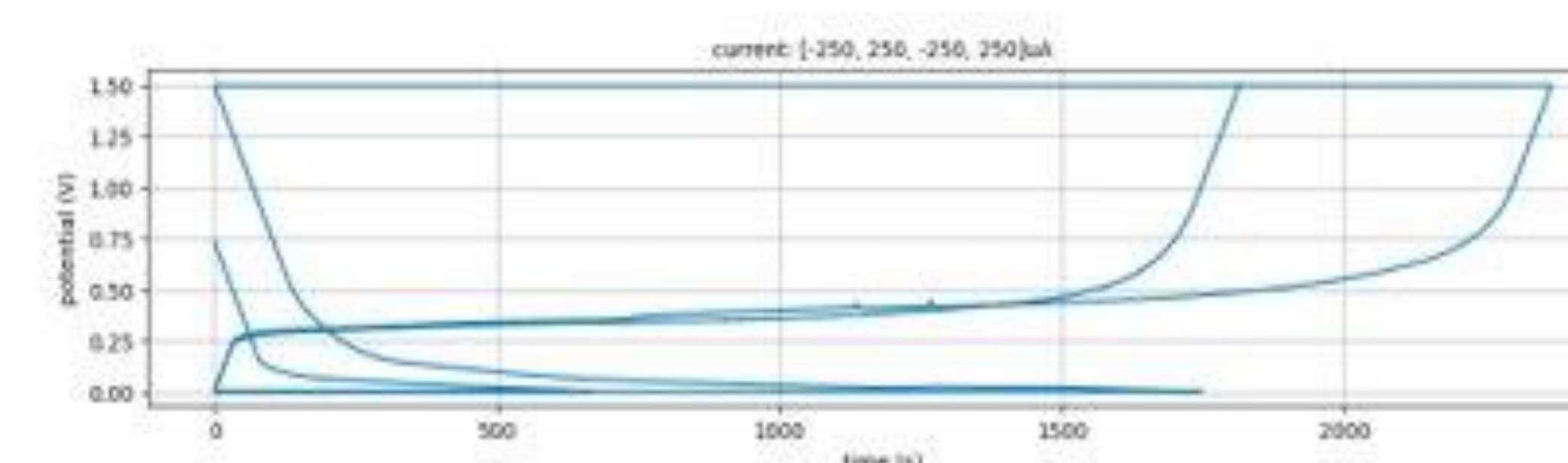
Charging and discharging full and/or half cell batteries through constant current methods.



Charging and discharging a lithium-ion full cell
•Lithium nickel manganese cobalt oxides (95%)



Cycling a lithium-ion half-cell, made in lab, >0.5 mAh with graphite.



CONCLUSION AND FUTURE WORK

I have implemented galvanostatic methods to the IO Rodeo Potentiostat with the implementation of a feedback loop that adjusts voltage given a set current. This method is useful in the analysis of electrochemical reactions and battery charging and discharging but is limited in precise qualification of electrochemical reactions.

The software limitations lie in the process of the feedback loop because it is an indirect system that mimics an Op-amp which directly converts current to voltage.

In future work, I plan to update the current software to implement live data reading. Currently the system only graphs until after the data is collected which is not beneficial in long tests and cycles where mistakes can happen early on.

Furthermore, I am working on a new program that will charge batteries at a constant current until a specified voltage is reached and it will then be held at a constant voltage till current is near zero which will indicate a full charge.

REFERENCES

Bard, A. *Electrochemical Methods Fundamentals and Applications*. Second Edition ed., John Wiley & Sons, 2001.

ACKNOWLEDGMENT

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